MaoMi + RANA

Fast tissue optics for microscopy

MaoMi generates microscopy movies from neural anatomy and activity. Its CUDA engine accelerates tissue optics; RANA models the local PSFs, polarization and exposure times that determine the recorded signal.

49.1×

faster optical calculation

NAOMi / MATLAB CPU	**179.644 s**
MaoMi / Standard CUDA	**3.662 s**

Full sampling, closely matching images

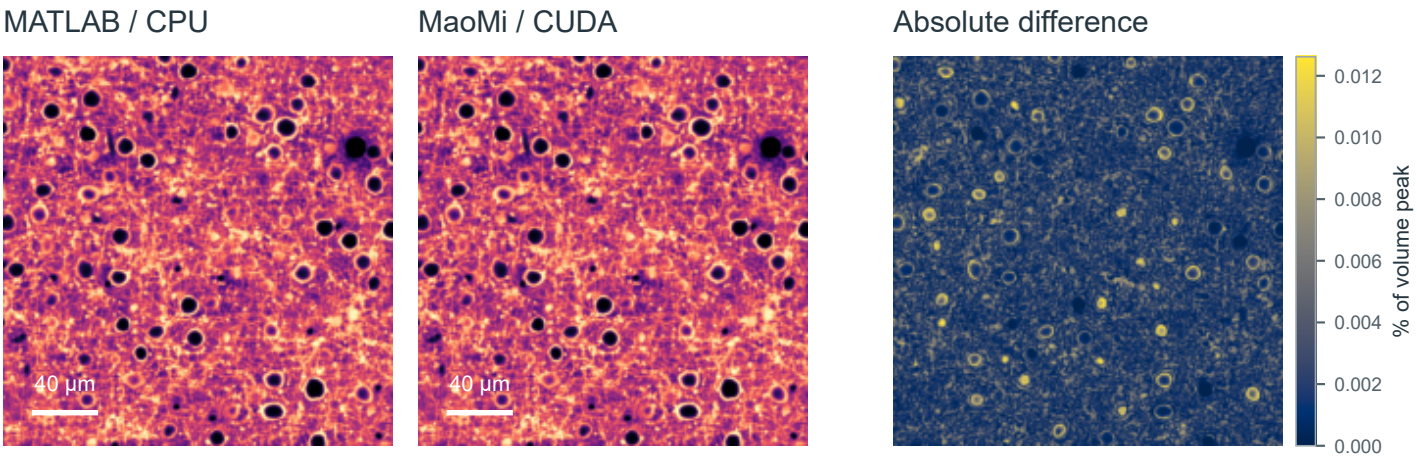


Figure 1. Plane 30 of the 512 × 512 × 60 volume. MATLAB and MaoMi share one linear display range; the difference panel uses the full reference-volume peak. Relative L2 difference over all voxels: **0.00667%**. Scale bars: 40 µm. [1]

WORKLOAD

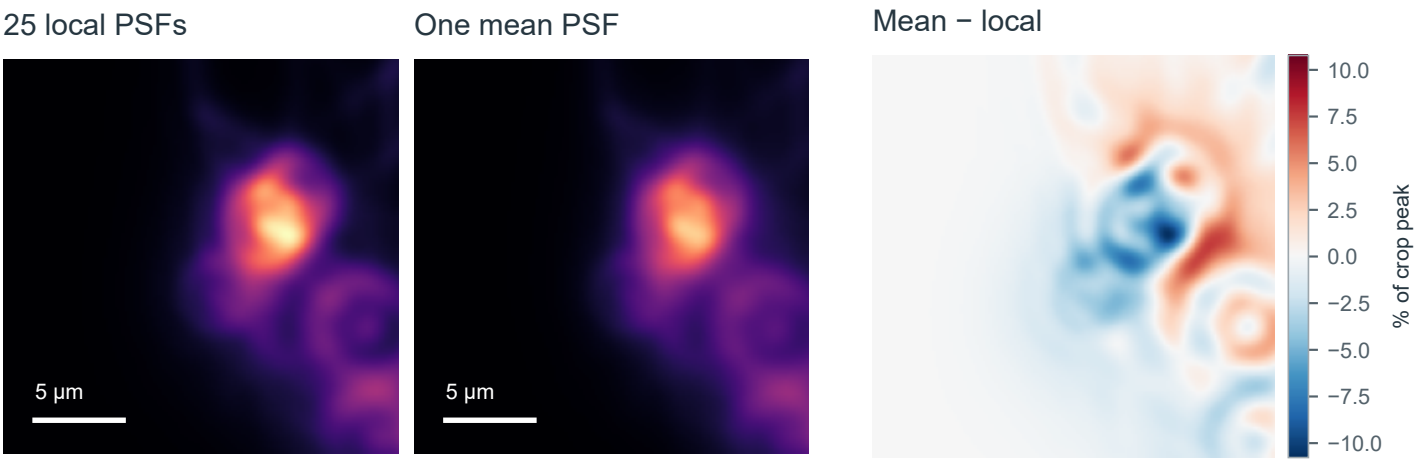
65 × 65 propagation sites, 0.5 µm voxels, identical saved tissue, scattering input and pupil. Three serial calls per route; medians on Ryzen 9 7950X and RTX 3080, CUDA batch 8.

TIMED OPERATIONS

Phase preparation, propagation, both optical tails and mask assembly. Input generation, static scanning, startup and file I/O are outside the timed stage. [1]

Where RANA changes the image

A field-dependent PSF changes shape and strength across the specimen. RANA applies those local kernels during image formation instead of replacing the field with a single average kernel.

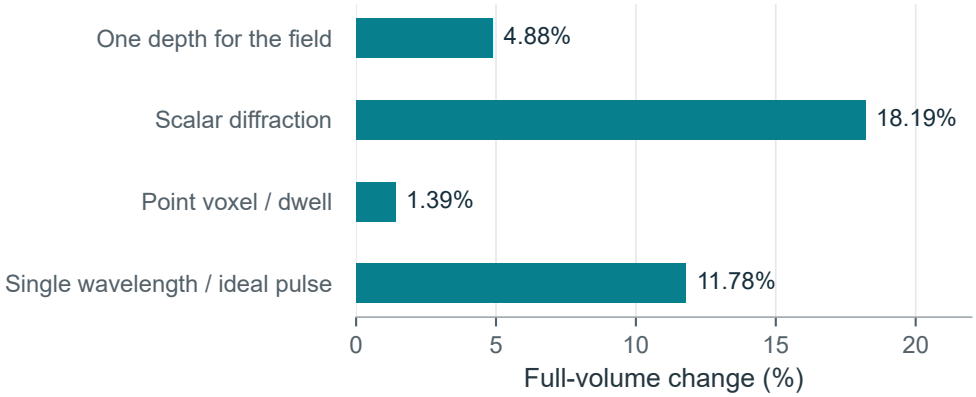


12.52%

Full-image change from averaging the PSF field. Both calculations use the same 25 HCIPy-generated kernels and cell signal. The difference comes from retaining their spatial variation. [3, 4]

Figure 2. A 22.4 µm crop from a 1024² photon image; selected by local signal brightness. The first two panels share a scale. The difference is mean minus local, relative to the crop peak. Scale bars: 5 µm.

Depth, high NA, finite sampling and pulse response



30.31%

volume change when all four simplifications are applied together.

0.1030%

difference from finer numerical quadrature.

Figure 3. MaoMi RANA reference: 256 × 256 × 16, NA 1.1, indices 1.38/1.33, depth 50–62 µm. PSF construction: 23.42 s CPU; rendering: 0.206 s CPU / 0.088 s CUDA. This experiment uses the standalone global cubic RBF renderer. [2]

From cell activity to camera counts

MaoMi accepts NAOMi tissue and activity files, custom cell recipes and external PSFs. Outputs include the latent specimen, expected signal, noisy images and cell labels for evaluating restoration and reconstruction.

01 Specimen
Tissue, cells, labels

>

02 Dynamics
Activity and motion

>

03 Exposure
Raster / global / rolling

>

04 Readout
Expected and noisy signal

A rolling shutter changes the shape of moving cells

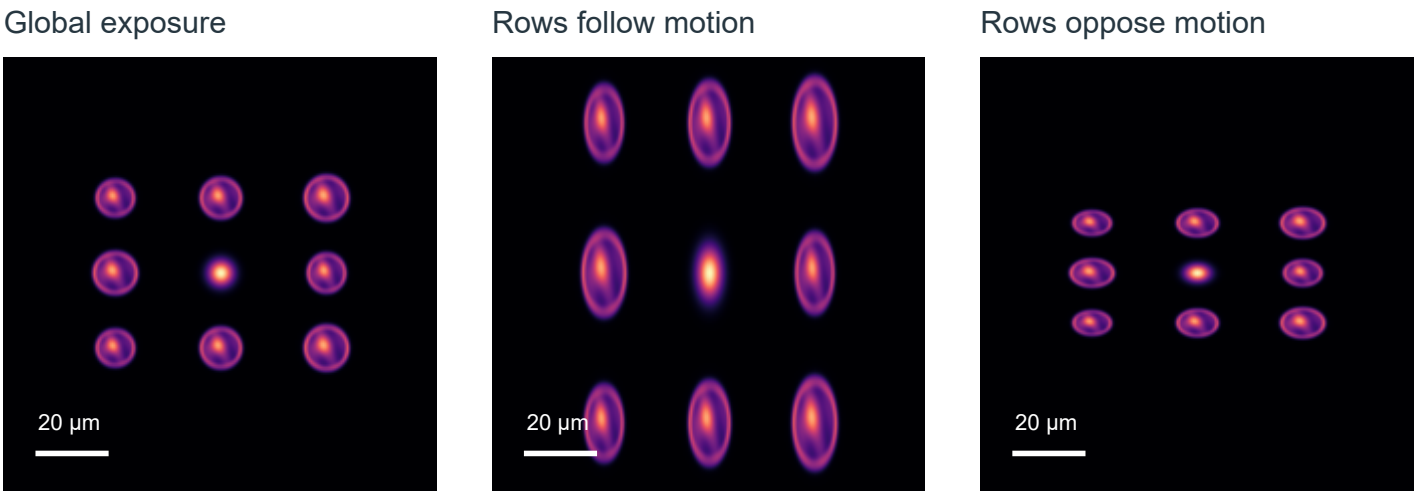


Figure 4. RANA-X acquisition study: sensor-plane motion at 3.2 mm/s, a 20 ms row sweep and 0.5 ms exposure. The calibration bead's RMS height is **2.003×** the global-exposure value when rows follow motion, and **0.666×** when they oppose it. Photon rates are integrated at each row's exposure times before detector noise. Scale bars: 20 µm. [3]

RANA-X biology examples

Calcium kinetics	Switch exponential and biexponential responses; compare fluorescence decay for fixed events.
Lifetime gates	Compare 0–1 ns and 2–4 ns gates for cells with equal total brightness but different decay times.
FRAP transport	Change diffusion from 0.15 to 0.80 µm²/s; measure recovery in a fixed 2 µm-radius region.

Heating changes the camera signal

In this RANA-X experiment, a simulated absorbing sample heats under illumination. Its temperature-dependent absorption changes the transmitted light and the camera signal.

532 nm illumination → absorption ↔ heat balance → camera charge

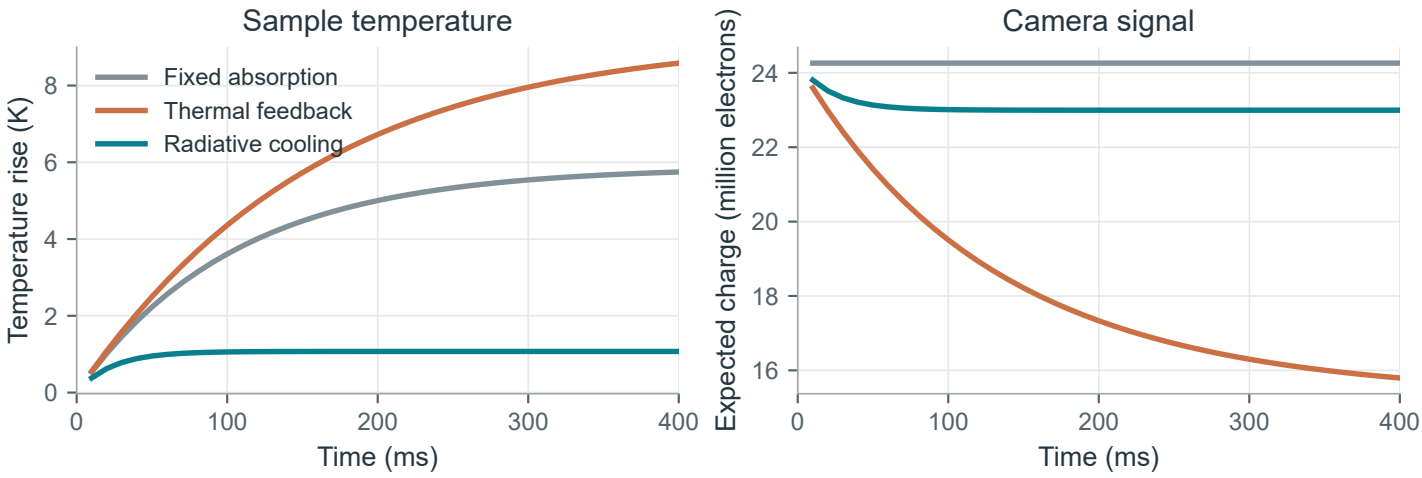


Figure 5. Fixed absorption gives a **5.75 K** rise at 0.4 s. Temperature-dependent absorption raises it to **8.58 K**; adding radiative cooling reduces it to **1.07 K**. Increased absorption lowers the transmitted-light signal. All three runs use the same illumination, acquisition and camera. [3]

Model settings. Six independent elements; 4×10^{12} incident photons/s per element; 40 steps of 10 ms; 300 K bath; optical depth 0.5; heat capacity 10^{-8} J/K; conductance 10^{-7} W/K. Feedback coefficient: 0 or 0.1 K^{-1}. Radiative case: emissivity 0.8, area 10^{-7} m². The external root solver and built-in iteration agree; resume reproduces all five output arrays exactly.

Software used. MaoMi 0.8.0b1 provides the tissue, activity, optics and acquisition features described here. The local-HCIPy, shutter, biology and material examples use the separate RANA-X 0.1.0.dev1 development code.

Sources and experiment records
[1] Song et al., NAOMi, J. Neurosci. Methods 358, 109173 (2021). MaoMi matched 4,225-site benchmark: dense_4225/summary.json, October 2026.
[2] MaoMi RANA comparison: equations, ablations, quadrature and construction/render costs.
[3] RANA-X records, 22–26 October: local_psf_1.npz; shutter_flow/results.json; B09_flim; B10_frap; thermal.json. Figure arrays and settings are retained with this revision.
[4] HCIPy documentation; HCIPy 0.7.1 supplied the kernels in Figure 2.